

Document AI Assistant — Project Report

1 Project Overview

The Document AI Assistant is an agent-based system designed to answer questions from documents using a combination of document processing, retrieval, analysis, and answer generation. The system consists of three main components: the Retriever Agent, Analyst Agent, and Answer Agent & Orchestrator.

2 Team Contributions

2.1 Zeina — Data & Retriever Agent

- Implemented the document ingestion pipeline, including loading, parsing, cleaning, chunking, and embedding.
- Integrated the vector database for document storage and retrieval.
- Implemented the Retriever Agent and its retrieval tools.
- Tested the implemented components thoroughly to ensure that the retrieval pipeline was working correctly.
- Provided the evidence bundle required by the other agents.

2.2 Alaa Khamees — Analyst Agent

- Implemented the Analyst Agent and its assigned tools, including calculation, data analysis, document comparison, and evidence-related functionality.
- The implemented part was not fully tested at the initial stage, which resulted in some errors during the integration of the complete project.

2.3 Alaa Abdelhakeem — Answer Agent & Orchestrator

- Implemented the Answer Agent and its formatting tools.
- Worked on integrating the different agents into the complete system.
- Helped resolve and rework parts of the Analyst Agent during integration.
- Performed the necessary testing and successfully ran the complete project.

3 Final Status

The three members contributed to their assigned components. During integration, some errors were identified in the Analyst Agent due to insufficient testing. These issues were addressed during the integration and testing phase, allowing the complete system to run successfully.

4 Application Screenshots

The following screenshots show the Document AI Assistant application in use, including document upload and ingestion, and the question-answering interface with citations and confidence scores.

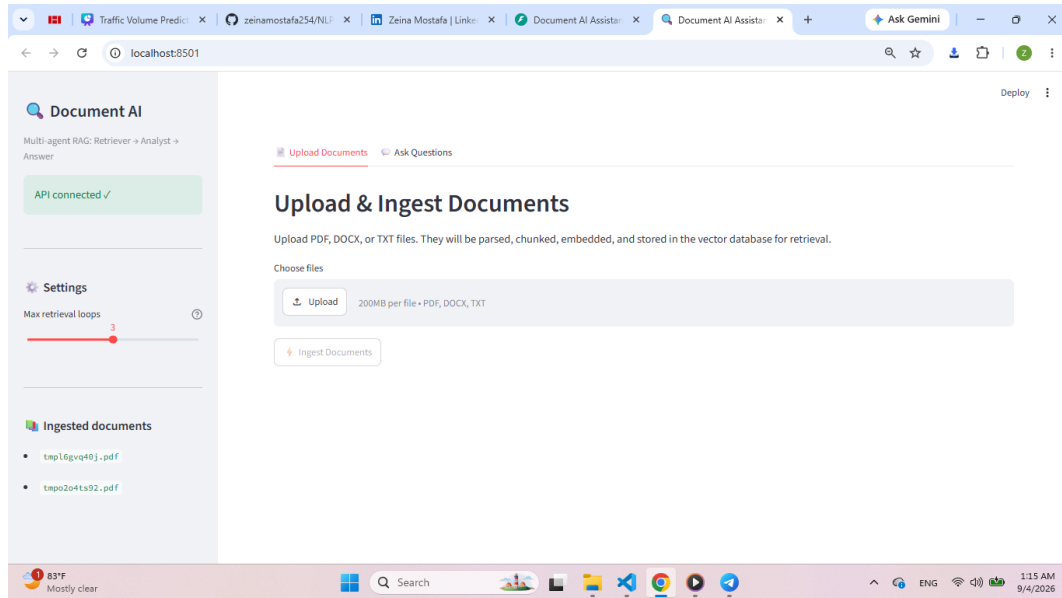


Figure 1: Upload & Ingest Documents interface with previously ingested files.

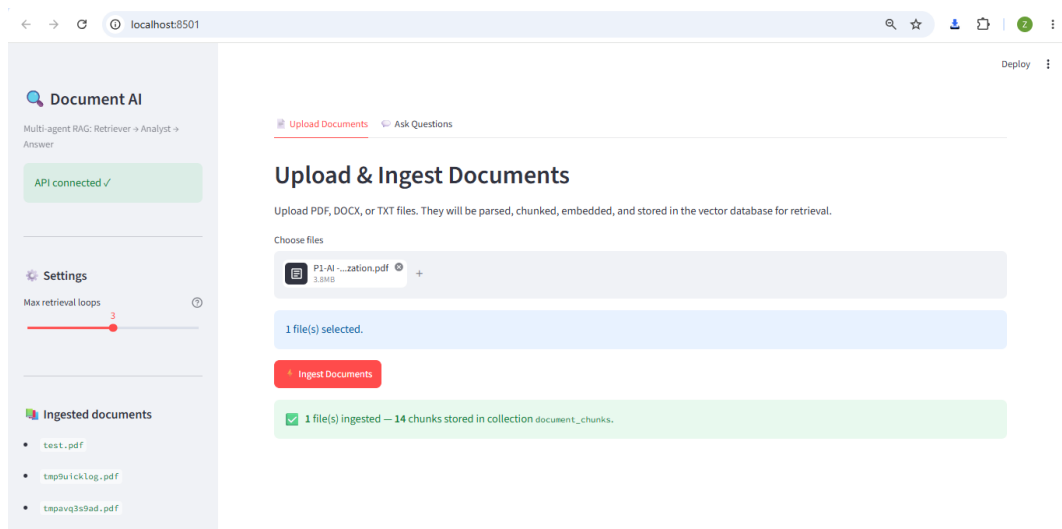


Figure 2: Document selected and successfully ingested, with chunks stored in the vector database.

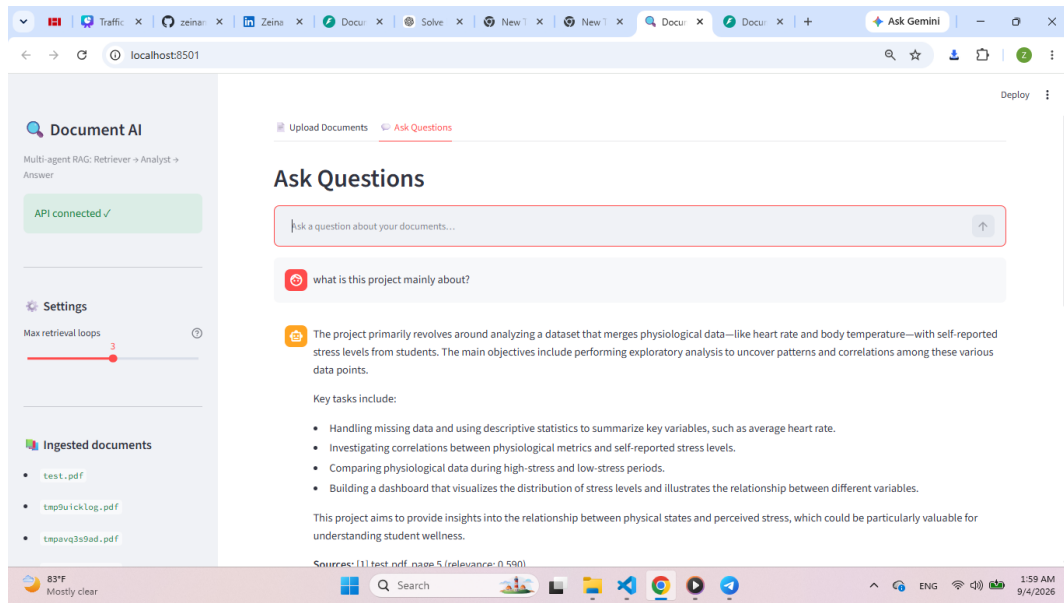


Figure 3: Ask Questions interface with a sample query and generated answer.

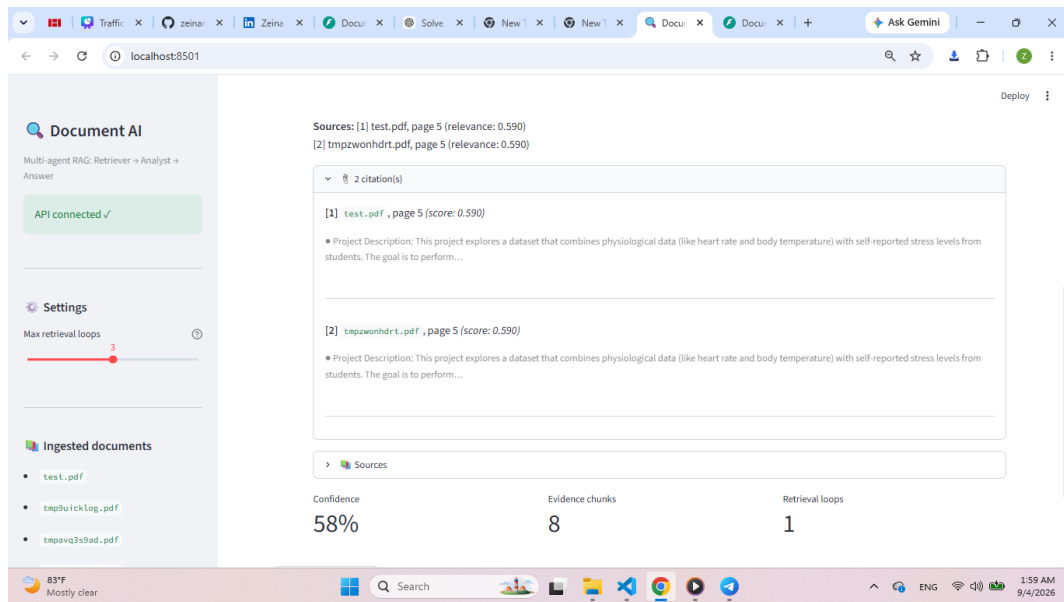


Figure 4: Citations, sources, and confidence metrics for the generated answer.

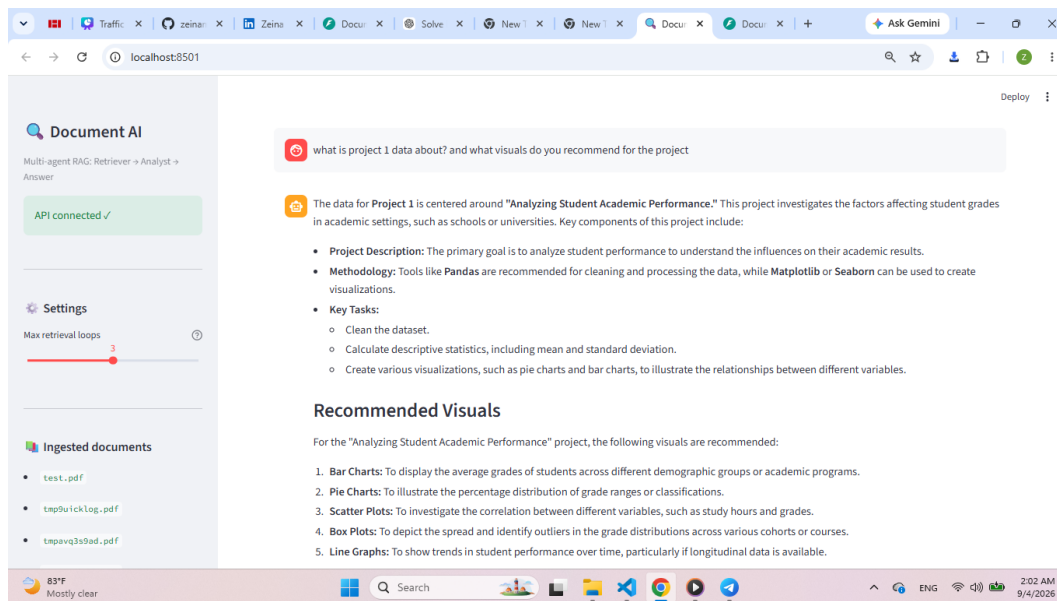


Figure 5: Example question with recommended visuals and key project details.

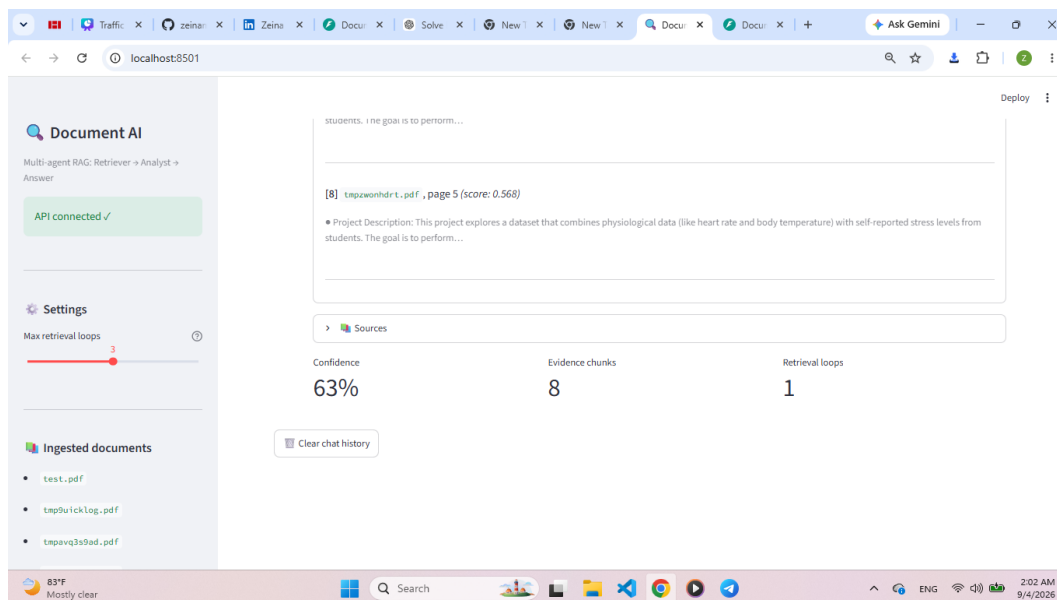


Figure 6: Evidence chunks, confidence score, and retrieval loop count for a query.

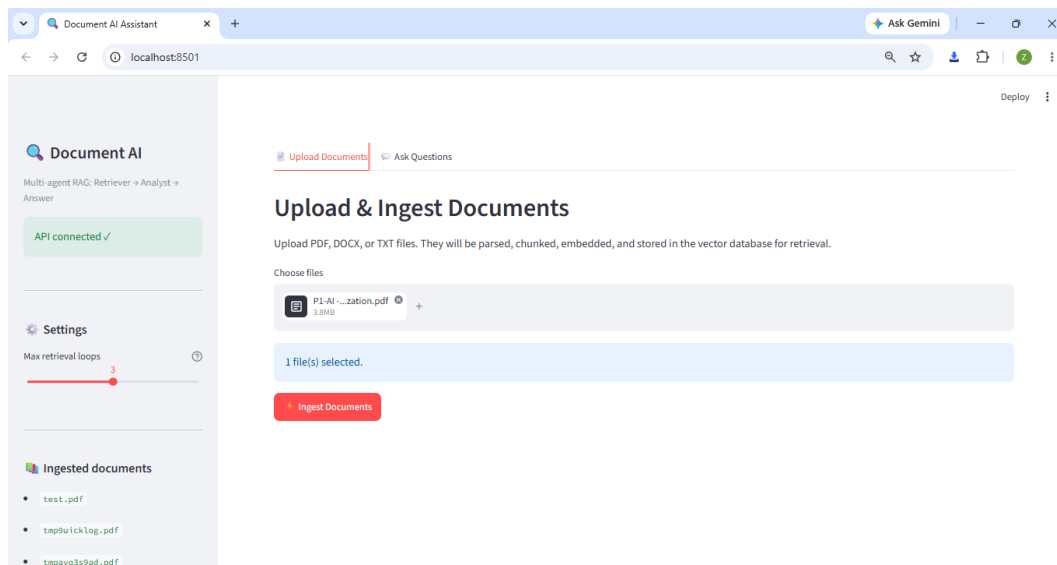


Figure 7: Document upload interface with a file ready for ingestion.